

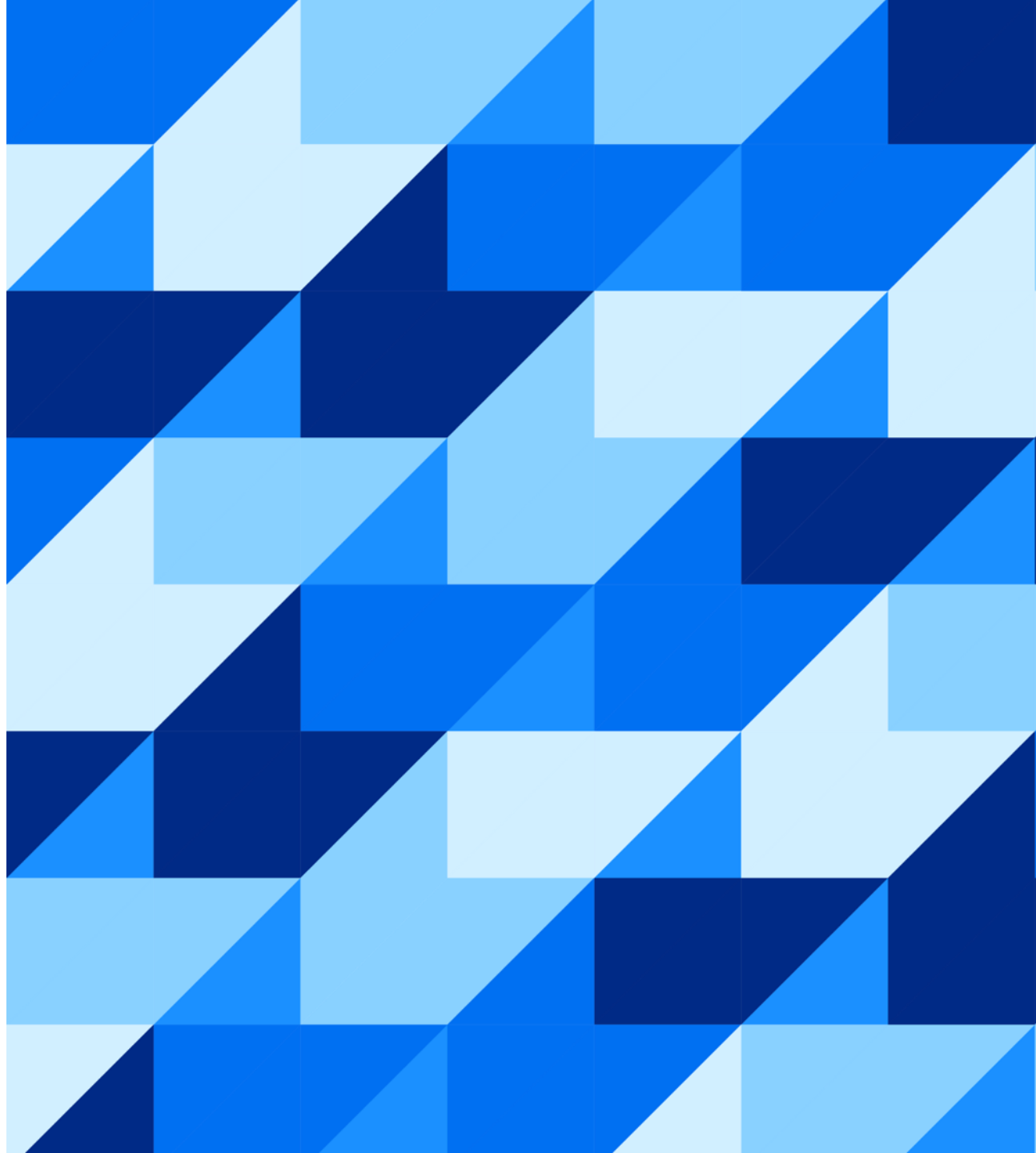


The ABAP strategy

towards ABAP Cloud development accelerated by ABAP AI

Fabian Fellhauer SAP SE
Boris Gebhardt, SAP SE
June, 2025

Public



Agenda

ABAP impact and strategy

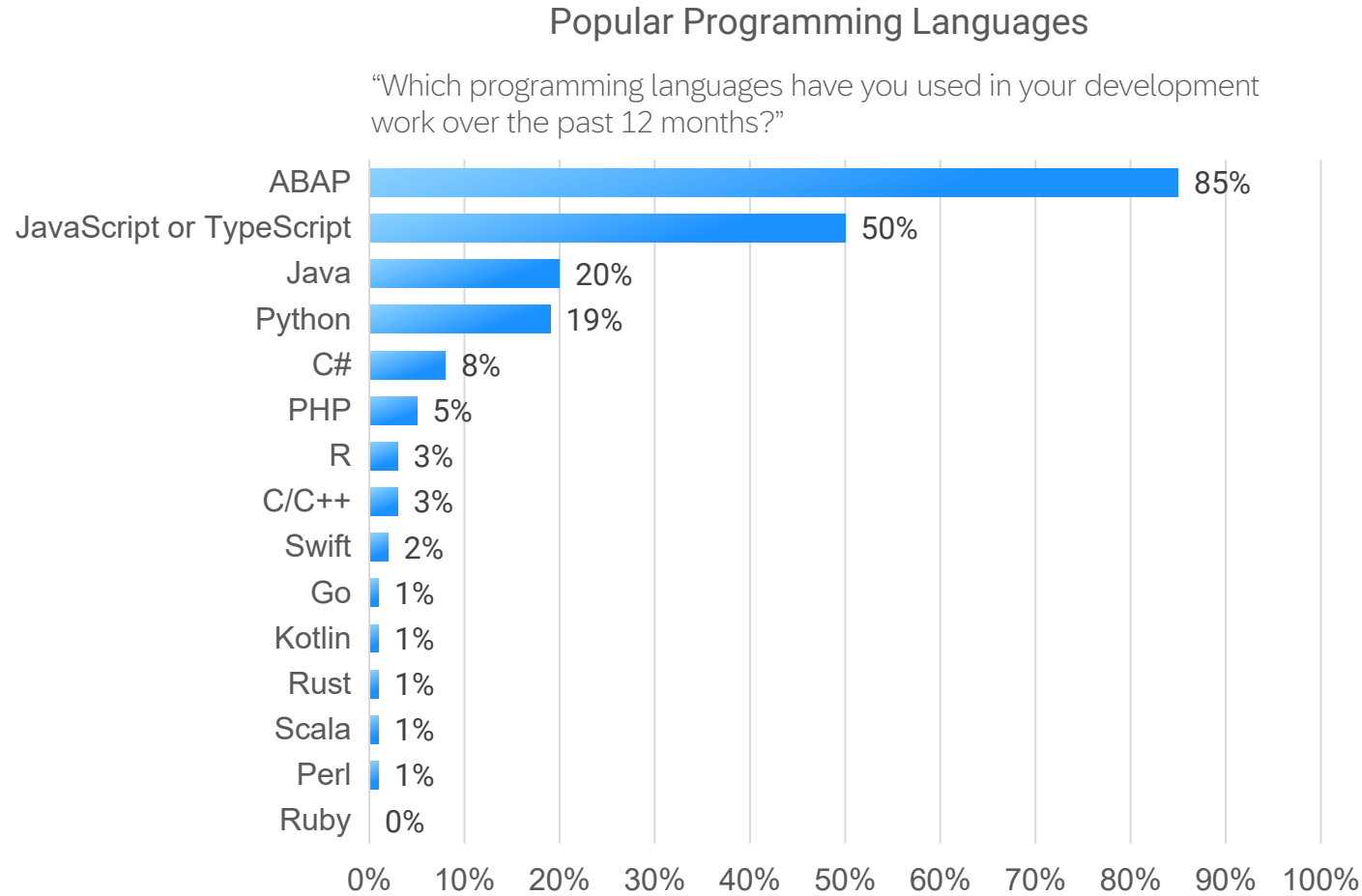
ABAP Cloud

ABAP AI

ABAP impact and strategy

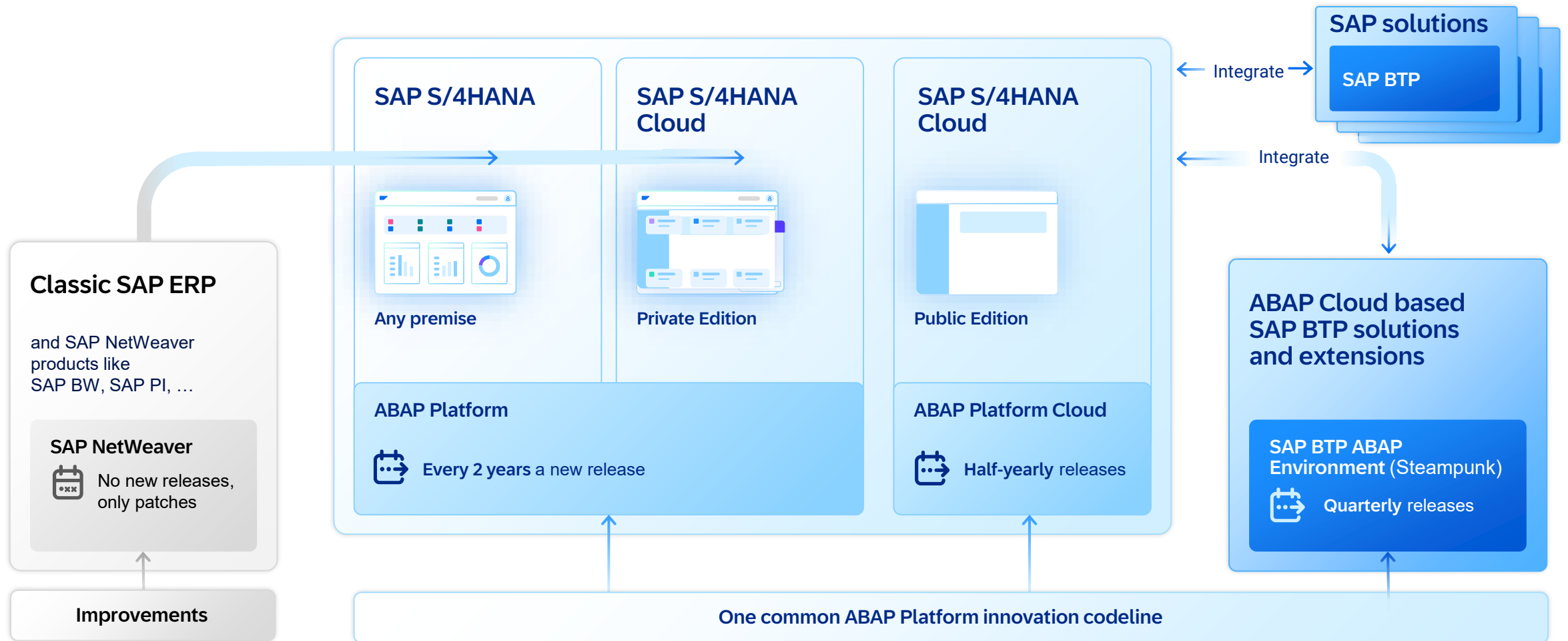


ABAP impact – 2024 SAP developer insight survey



[The 2024 Developer Insights Survey: The Report - SAP Community](#)

ABAP as main technology provider for SAP S/4HANA



SAP BTP ABAP Environment (Steampunk)

Frontrunner for cloud and innovations

ABAP Cloud development model

- Upgrade stable
- Optimized for cloud operations
- SAP S/4HANA optimized



SAP BTP ABAP Environment (Steampunk)

ABAP Cloud development model

Steampunk infrastructure

SAP BTP services

Gardener (K8S), SAP HANA Cloud

Steampunk infrastructure

- Scalable
- Resilient
- Automated operation



ABAP Cloud





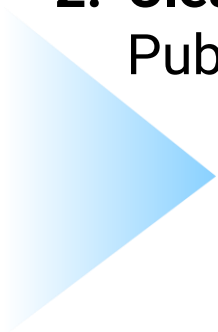
ABAP Cloud combines

1. Cloud Technologies

ABAP Cloud Language, ABAP RESTful Application Programming Model (RAP), SAP Fiori UI Technology, Integration technologies, technical services, business services, ABAP Development Tools, Gen AI powered development,...

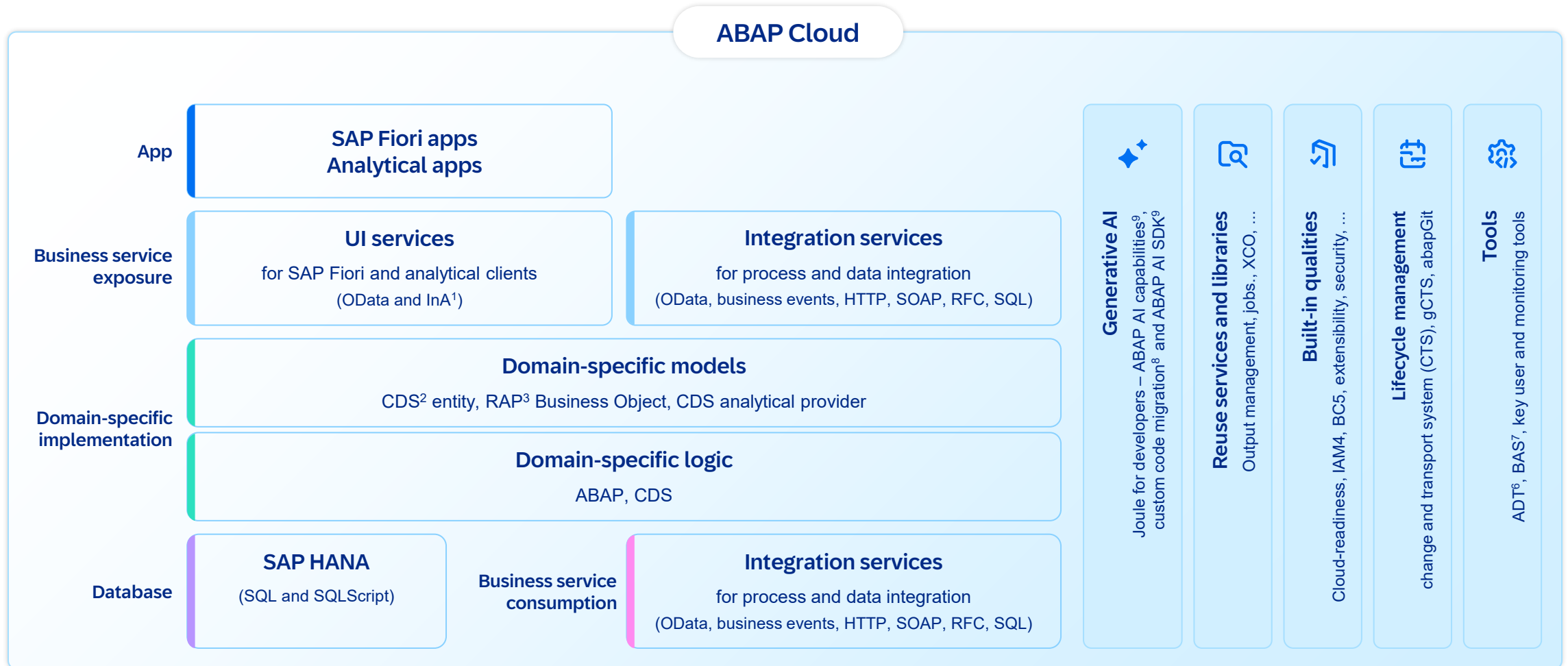
2. Clear Separation between SAP code and custom code

Public extension points, local APIs, remote APIs



Cloud-Ready and Clean Core by Default

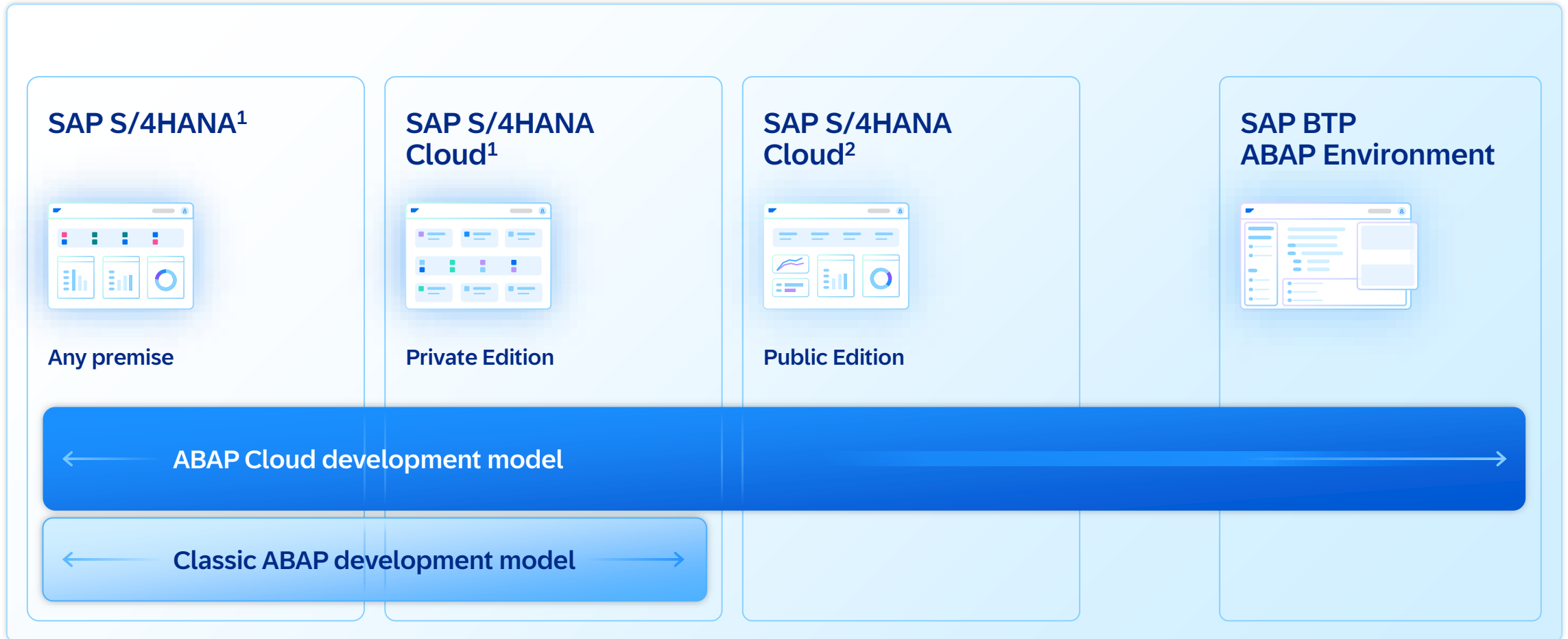
ABAP Cloud map



1 Information access / 2 Core Data Services / 3 ABAP RESTful application programming model / 4 Identity & Access Management / 5 Business Configuration / 6 ABAP Development Tools /

7 Business Application Studio / 8 Planned to be available in future releases / 9 Available for SAP S/4HANA Cloud Public Edition and SAP BTP ABAP environment

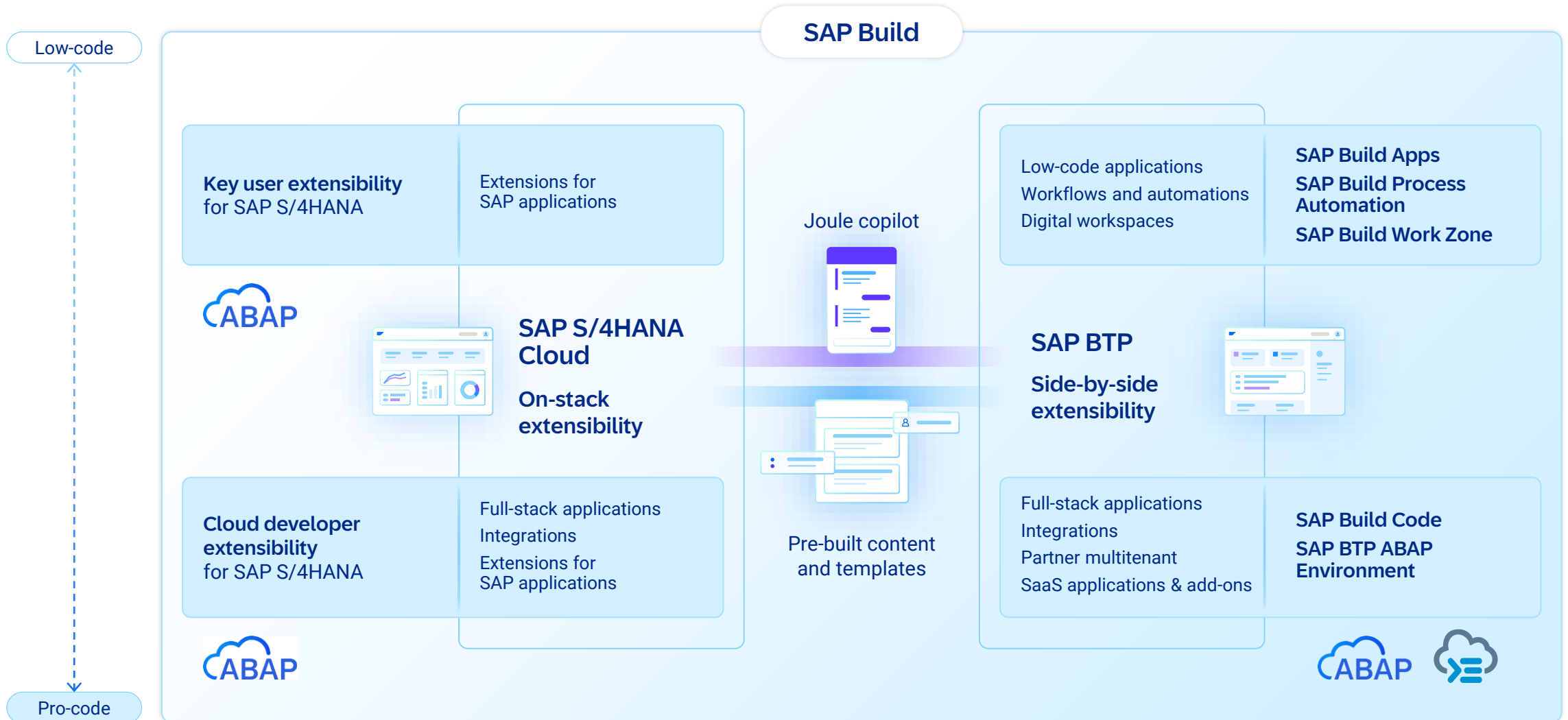
ABAP Cloud – One development model for SAP S/4HANA and SAP BTP



¹ SAP S/4HANA any premise or SAP S/4HANA Cloud, private edition release ≥ 2022

² SAP S/4HANA Cloud, public edition release ≥ 2208, 3-system landscape required

ABAP Cloud in SAP Build as essential part of SAP's development portfolio



ABAP Cloud establishes the industry standard for developing enterprise business applications

We accompany the journey.

Increase Developer Productivity

with generators, upskilling, migration tools

We ensure competitiveness.

Incorporate Emerging Technologies

like generative AI, Fiori, HANA

We are reliable.

Grant Cloud-Native Qualities Out-Of-The Box

like resilience, performance, observability, scalability, extensibility, testability

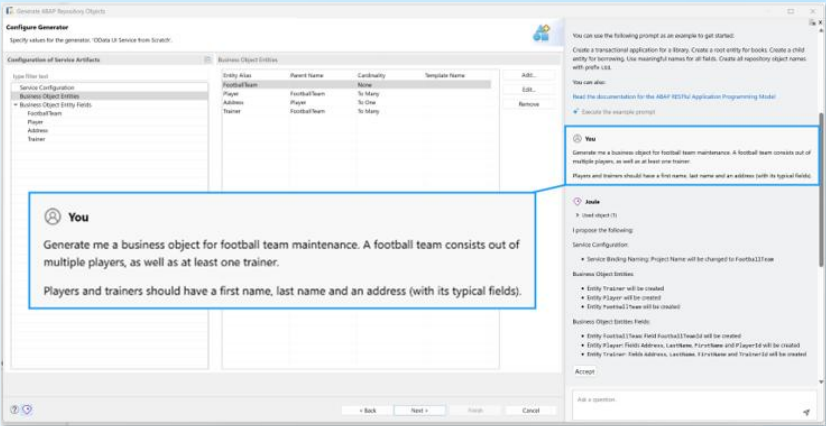
NOW
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NEXT
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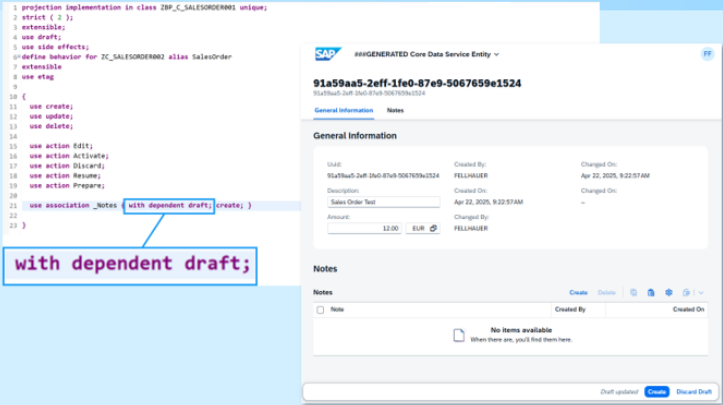
SOON
(later)

INNOVATE YOUR ENTERPRISE

RAP Business Object Generator
Generate business objects with AI assistance



Static Draft Scope
Bundle up different business objects



Analytical Table
Aggregation and grouping capabilities

☐	Document Number	Description	Posting Key	Document Type	Posting Date	Document Amount	Currency	Offset Account	Assignment
Assignment 2938481									
☐	90000201	Laptops	40	Invoice	11.21.2020	3.100.90	EUR	660200	2938481
☐	90000213	Servers	40	Invoice	11.21.2020	11.932.71	EUR	700000	2938481
☐	90000183	Tables	40	Invoice	11.21.2020	39.392.74	EUR	700000	2938481
☐	90000623	Printers	40	Invoice	11.21.2020	9.349.17	EUR	660200	2938481
☐	90000871	3D Printers	40	Invoice	11.21.2020	1.283.57	EUR	550000	2938481
						65,056.09	EUR		
Assignment 29392182									
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☐	90003421	3D Printers	50	Invoice	11.21.2020	6.192.00	EUR	700000	2938481
☐	90002125	Servers	40	Invoice	11.21.2020	12.342.00	EUR	700000	2938481
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Assignment 29392182									
☐	90000201	Phone	40	Invoice	01.14.2021	11.193.70	EUR	660200	2938481
☐	90000213	Servers	40	Invoice	01.14.2021	2.932.71	EUR	700000	2938481
☐	90000183	Chair	40	Invoice	01.14.2021	11.193.70	EUR	700000	2938481
☐	90000623	Printers	40	Invoice	01.14.2021	5.349.17	EUR	660200	2938481
☐	90000871	3D Printers	40	Invoice	01.14.2021	2.283.57	EUR	550000	2938481
						33,151.89	EUR		
						148,989.99	EUR		

BOOST YOUR EFFICIENCY

¹ See <https://help.sap.com/docs/abap-cross-product/roadmap-info/abap-cloud-roadmap-information> for further details. Mentioned releases may be subjected to be changed without further notice.

NOW
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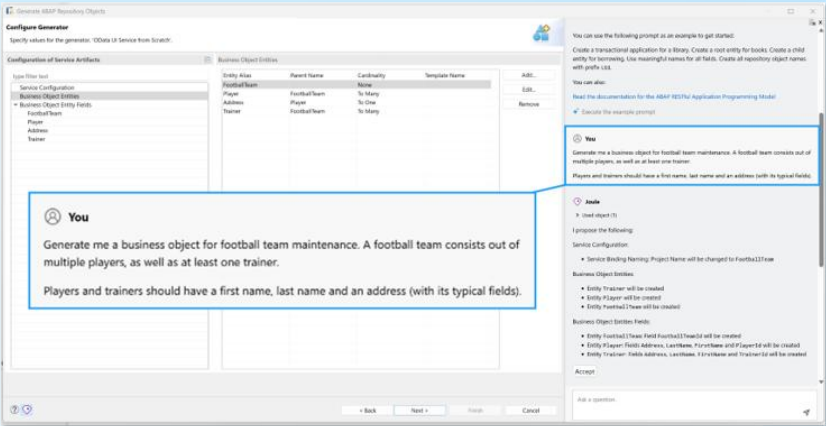
NEXT
(2508)

SOON
(later)

INNOVATE YOUR ENTERPRISE

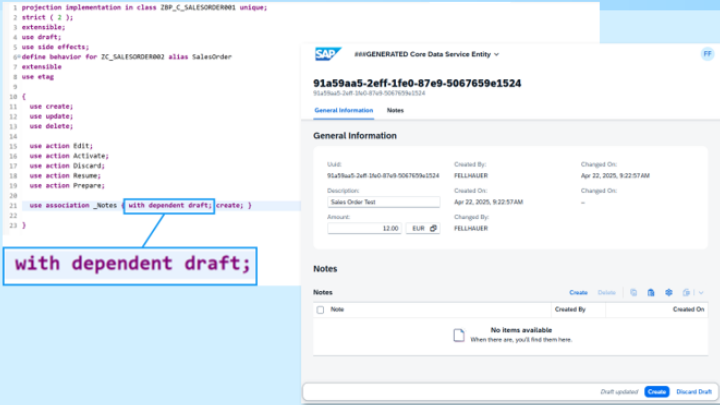
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	90000623	Printers	40	Invoice	11.21.2020	9,349.17	EUR	660200	2938481
	90000871	3D Printers	40	Invoice	11.21.2020	1,282.57	EUR	550000	2938481
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Assignment 29392182									
	90042121	Laptops	50	Invoice	11.21.2020	4,935.71	EUR	660200	2938481
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	90000871	3D Printers	40	Invoice	01.14.2021	2,283.57	EUR	550000	2938481
						32,151.89	EUR		
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BOOST YOUR EFFICIENCY

¹ See <https://help.sap.com/docs/abap-cross-product/roadmap-info/abap-cloud-roadmap-information> for further details. Mentioned releases may be subjected to be changed without further notice.

ABAP Cloud Roadmap ¹



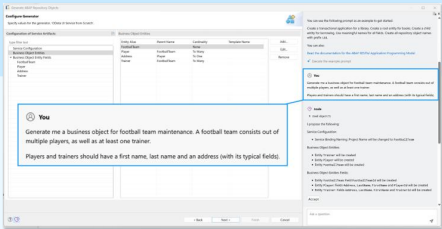
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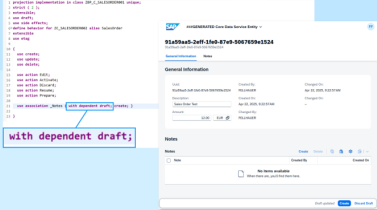
SOON
(later)

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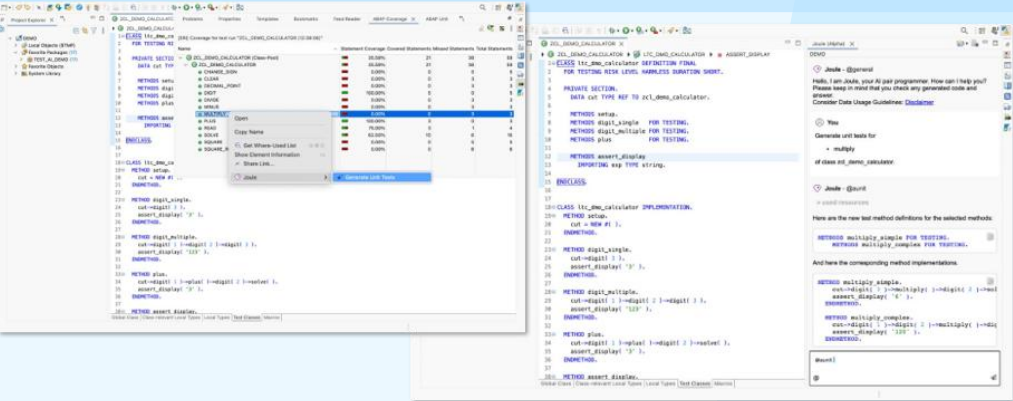


Analytical Table
Aggregation and grouping capabilities

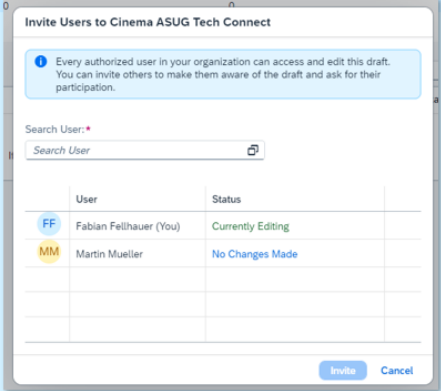
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Assignment 2508242								
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BOOST YOUR EFFICIENCY

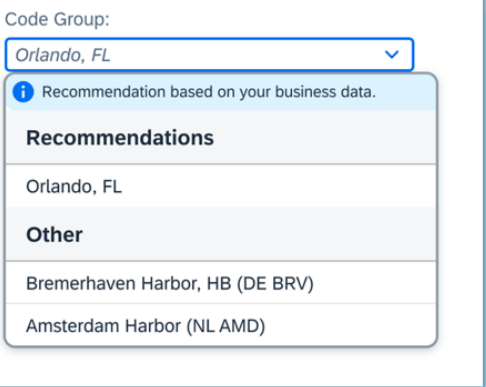
Joule Developer Capabilities for ABAP
explain & predict coding, generate unit tests and much more...



Collaborative Draft
Working together on the same draft instance



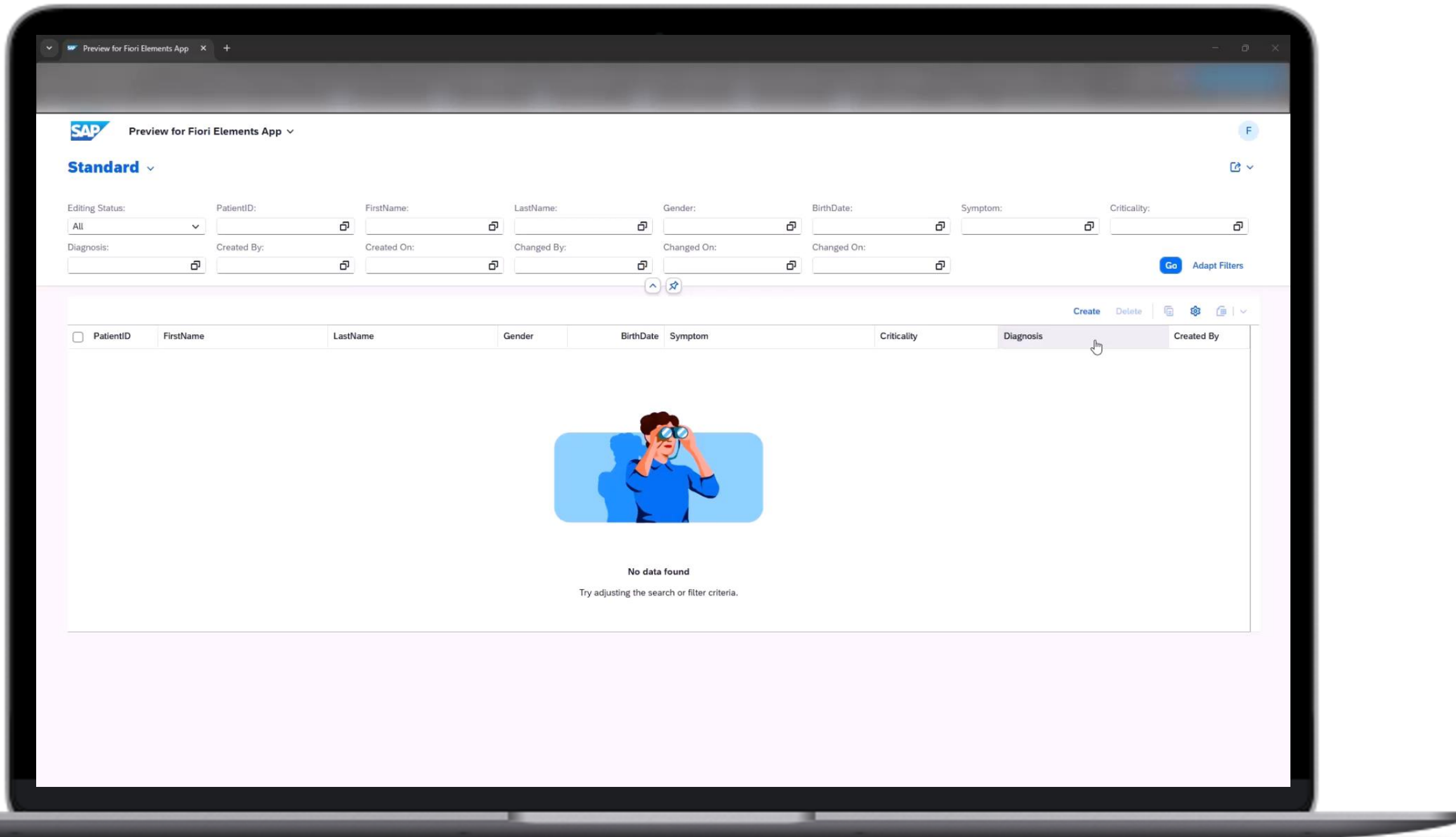
Recommendations
AI-based input recommendations



¹ See <https://help.sap.com/docs/abap-cross-product/roadmap-info/abap-cloud-roadmap-information> for further details. Mentioned releases may be subjected to be changed without further notice.

Recommendations

AI-based input recommendations



ABAP Cloud Roadmap ¹



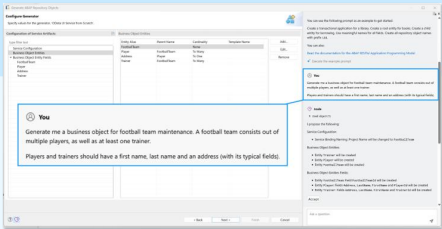
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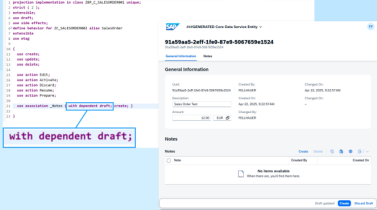
SOON
(later)

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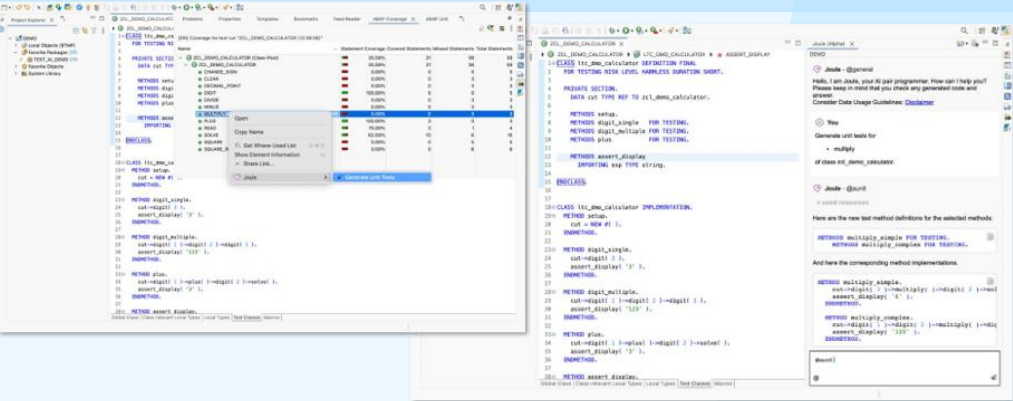


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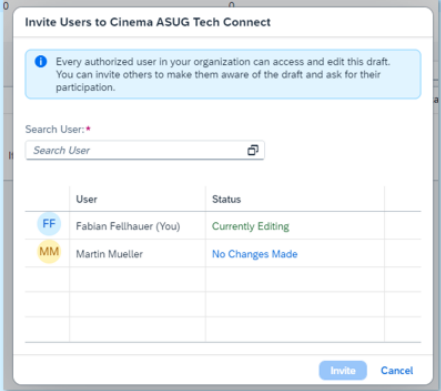
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BOOST YOUR EFFICIENCY

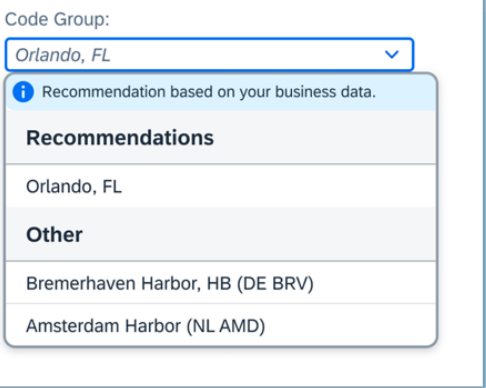
Joule Developer Capabilities for ABAP
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Collaborative Draft
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AI-based input recommendations



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ABAP AI strategy



ABAP Use Cases for Generative AI

Accelerate

Joule for developers

Integrated in ABAP Development Tools

Empower

ABAP AI SDK powered by ISLM

ABAP Client for SAP BTP GenAI Hub

Transform

ECC → SAP S/4HANA → Clean Core and ABAP Cloud

Integrated in the ABAP Test Cockpit and the Custom Code Migration App

Use Case: Accelerate*

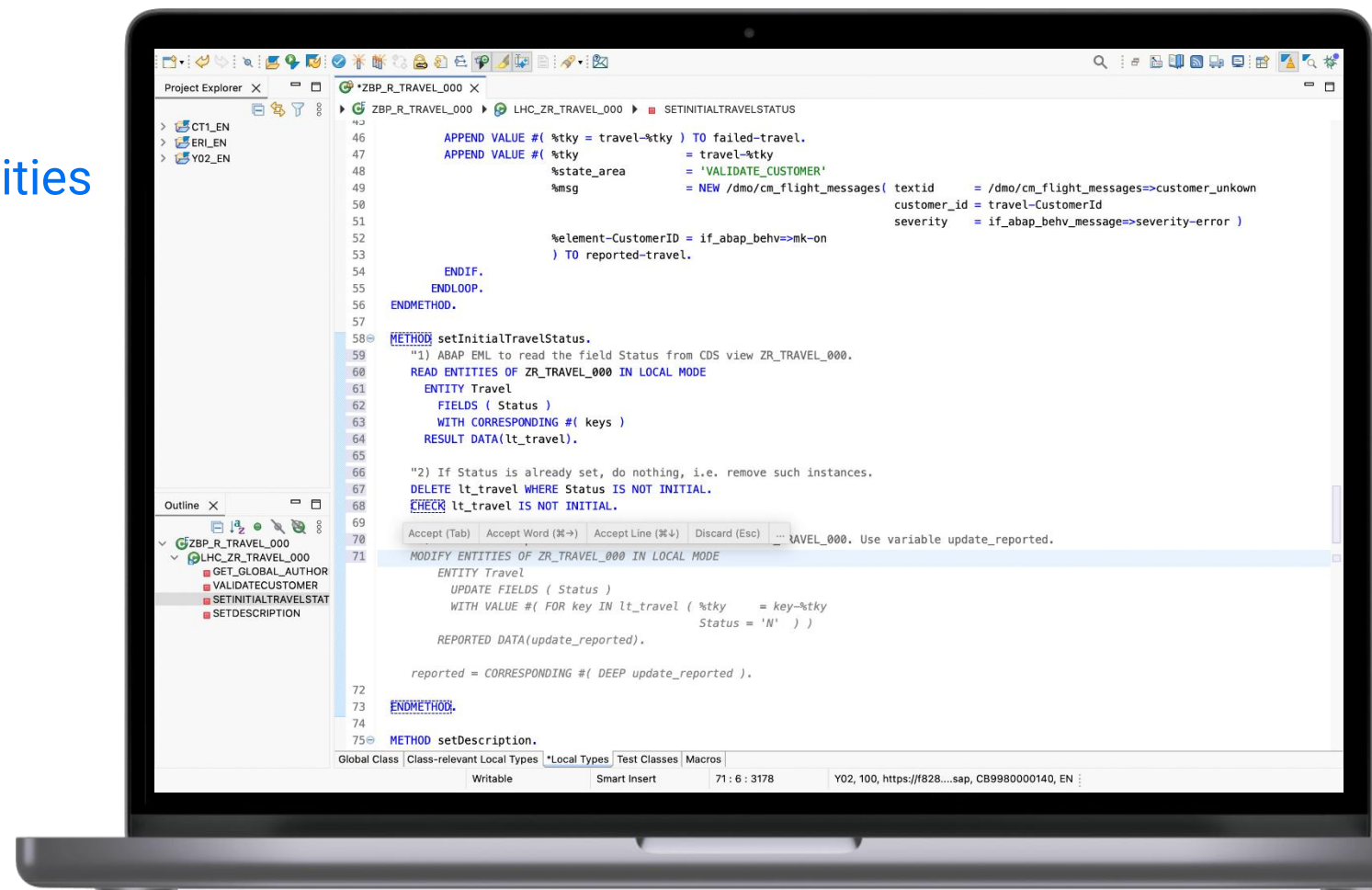
Joule for developers, ABAP AI capabilities

Scope for GA within 2502

- Joule Copilot
- Predictive Code Completion
- ABAP Code Explain
- ABAP CDS Explain
- ABAP Unit Test Generation
- CDS Unit Test Generation

Products*

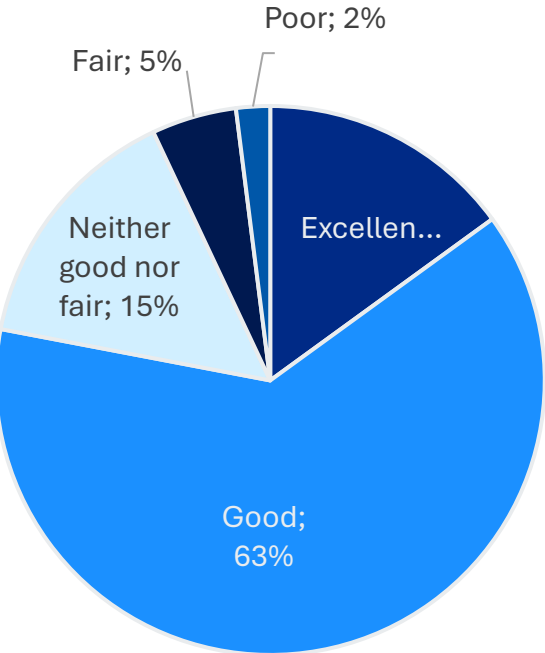
- SAP BTP ABAP Environment as of release 2502
- SAP S/4HANA Cloud Public Edition as of release 2502
- SAP S/4HANA Cloud Private Edition as of release 2025.



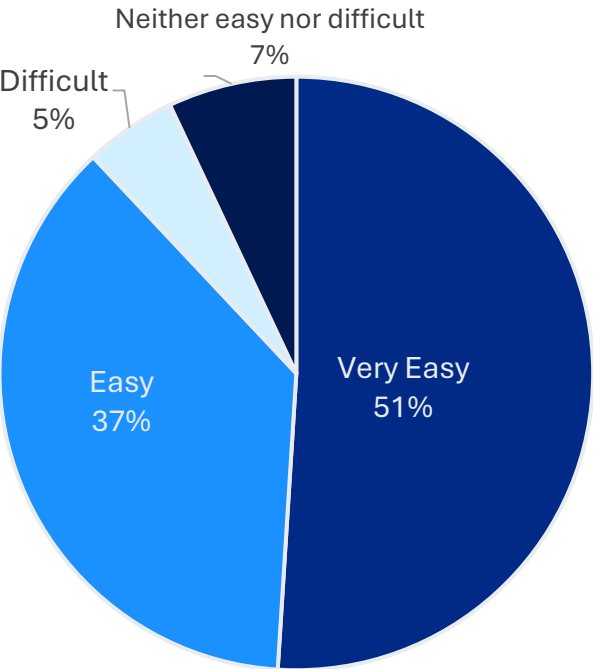
Early Feedback Joule for developers, ABAP AI capabilities

Internal developers + customer testing

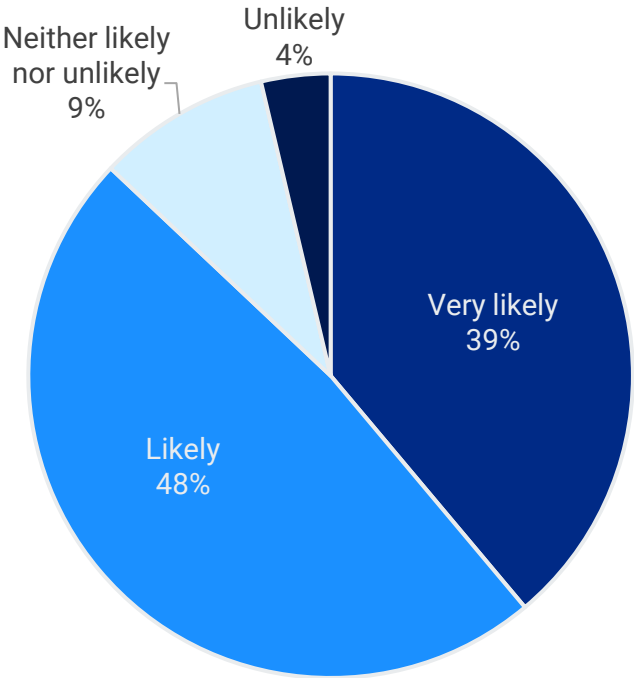
UX



Ease of Use



How likely will you recommend it?



Use Case: Empower*

Empower

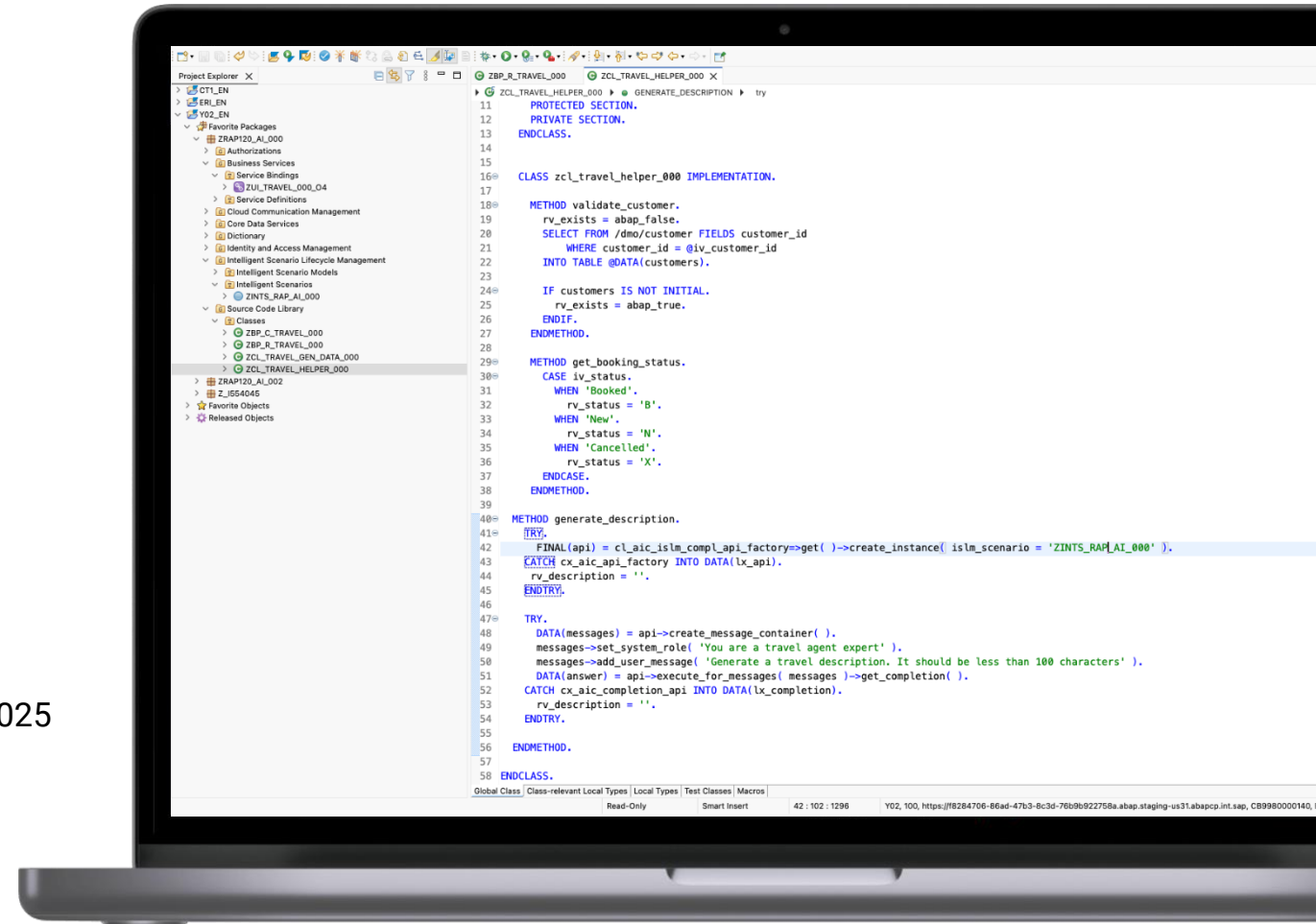
GA within 2502: ABAP AI SDK powered by Intelligent Scenario Lifecycle Management

- Central, standardized access to generative AI hub in SAP AI Core
- Used by SAP for ABAP AI functionality
- Easy to configure, easy to consume

Products*

- SAP BTP ABAP Environment as of release 2502
- SAP S/4HANA Cloud Public Edition as of release 2502
- SAP S/4HANA Cloud Private Edition and OnPrem as of release 2025 via standard delivery.**

** Lower releases of SAP S/4HANA Cloud Private Edition and OnPrem are supported via Transport-based Correction Instructions (TCI), see details in [SAP Note 3513374](#)



Use Case: Transform*

Transform

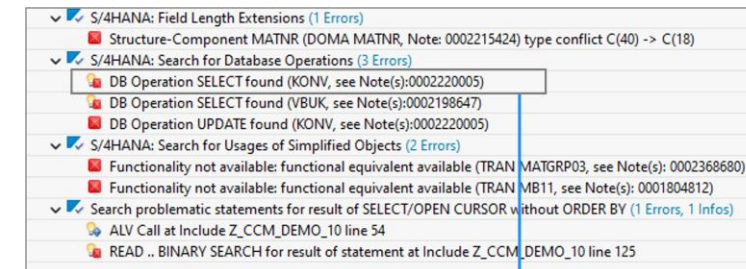
Custom Code Migration to SAP S/4HANA

- Embedded in ABAP development tools for Eclipse
- Get AI-based explanation of legacy code (also planned in Custom Code Migration app)
- Get AI-based explanation of SAP S/4HANA related ATC findings and Simplification Items
- Get AI-based proposal for code adaptation

Custom Code Migration to ABAP Cloud

- Embedded in ABAP Development Tools for Eclipse
- Get AI-driven code suggestions for adaptation to usage of SAP APIs

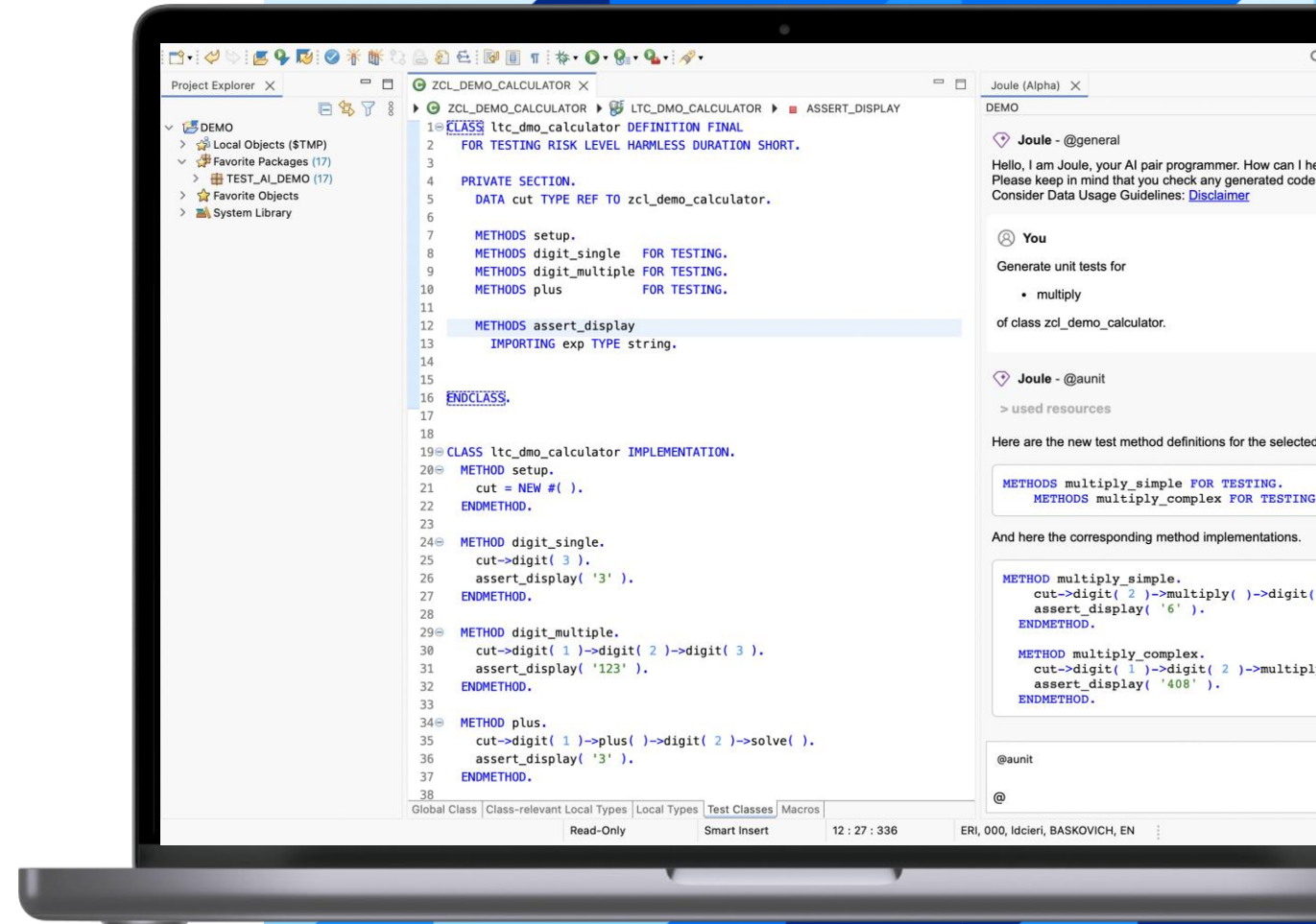
Example: Explain ATC finding and get AI code proposal for SAP S/4HANA adaptation



* This presentation and SAP's strategy and possible future developments are subject to change and may be changed by SAP at any time for any reason without notice.

Demo

Joule capabilities in ABAP



RAP Business Object Generator*

The screenshot displays the 'Generate ABAP Repository Objects' window, specifically the 'Configure Generator' tab. The interface is divided into several sections:

- Configuration of Service Artifacts:** A tree view on the left shows the hierarchy: Service Configuration > Business Object Entities > Business Object Entity Fields. The 'Business Object Entity Fields' section is expanded, showing 'FootballTeam', 'Player', 'Address', and 'Trainer'.
- Business Object Entities Table:** A table with columns: Entity Alias, Parent Name, Cardinality, and Template Name. It lists the following entities:

Entity Alias	Parent Name	Cardinality	Template Name
FootballTeam		None	
Player	FootballTeam	To Many	
Address	Player	To One	
Trainer	FootballTeam	To Many	
- AI Prompts and Responses:** On the right, there are two chat windows. The first, labeled 'You', contains a prompt: 'Generate me a business object for football team maintenance. A football team consists out of multiple players, as well as at least one trainer. Players and trainers should have a first name, last name and an address (with its typical fields).' The second, labeled 'Joule', provides a response detailing the proposed service configuration and the specific entities and fields to be created.

At the bottom of the window, there are navigation buttons: '< Back', 'Next >', 'Finish', and 'Cancel'.

Takeaways



Takeaways

ABAP Cloud

- Combines cloud technologies and clear separation of SAP and customer/partner code
- Default for any new ABAP development on SAP S/4HANA and on SAP BTP
- Is a key pillar of the SAP S/4HANA Extensibility strategy
- Used productively by customers and partners in SAP S/4HANA and on SAP BTP



Generative AI for ABAP

- 3 main scenarios: Accelerate, Empower, Transform
- Accelerate: GA as of 2502 in SAP BTP ABAP Environment
SAP S/4HANA Cloud Public Edition
- Empower: GA as of 2502 in the cloud and downports until 2021 for SAP S/4HANA OnPrem and Private Cloud
- Transform: Planned for 2025



Become an ABAP Cloud certified developer

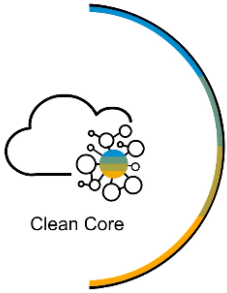
SAP Learning journeys valid for all SAP S/4HANA editions



Learn | Get certified | My learning | Explore SAP | 

Practicing Clean Core Extensibility For SAP S/4HANA Cloud

5 Units | 8 hrs | Free until certification



Clean Core



Up to Date



Cloud Compliant Extensions and Integration



Optimal Master Data

Overview

This learning journey will show you how to implement extensions for SAP S/4HANA Cloud consistent with clean core principles and aligned with your business use cases.

Learning objectives

After completing this learning journey, the learner will be able to build custom extensions in SAP S/4HANA Cloud using various tools and based on the concepts of consumer-grade user experience (UX), clean core, the SAP S/4HANA Cloud Extensibility Model and the ABAP Cloud Development Model.

Your current experience in this topic

BEGINNER | INTERMEDIATE | ADVANCED

Prerequisites

- [Learning the Basics of ABAP Programming on SAP BTP](#)
- [Acquiring Core ABAP Skills](#)

On this page: [Course Content](#) | [SAP Learning Group](#) | [Certification](#)



Prepare for certification

Learn new skills you need at your own pace with our bite-sized lessons.



 SAP Learning Journey: [Practicing Clean Core Extensibility for SAP S/4HANA Cloud](#)

 Become an [SAP Certified Associate – ABAP Cloud Back-End Developer](#)



Further information

- [ABAP Extensibility Guide](#) | SAP Community
 - [ABAP Cloud Development Guide](#) | SAP Help Portal
 - [ABAP RESTful Application Programming Model \(RAP\)](#) | SAP Community, SAP Help Portal
 - [ABAP Roadmap Information](#) | For ABAP-based Products and ABAP Cloud
 - [ABAP Cloud sessions at Devtoberfest by SAP TechEd in 2024](#) | [ABAP Cloud sessions at SAP TechEd 2024](#) | Blog posts (incl. replays)
-

Public SAP Web sites

ABAP Development Community: www.sap.com/community/topic/abap

ABAP Extensibility Community: <https://pages.community.sap.com/topics/abap-extensibility>

ABAP Testing and Analysis Community: <https://community.sap.com/topics/abap-testing-analysis>

SAP BTP ABAP Environment Community: <https://community.sap.com/topics/btp-abap-environment>

SAP S/4HANA Cloud ABAP Environment Community: <https://community.sap.com/topics/s4hana-cloud-abap-environment>

SAP training and certification opportunities

www.sap.com/education – e.g. trainings S4D400, S4D430, S4D437, and HA400.

<https://learning.sap.com/> – e.g., search for ABAP or ABAP Cloud

[SAP Certified Associate – ABAP Cloud Back-End Developer](#)

Thank you.

Contact information:

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